Competent
38/45
85%

TOOLS, EQUIPMENT AND MATERIALS

SUMMATIVE ASSESMENT

Module #	671101000-KM-02
NQF Level	level 4
Notional hours	80
Credit(s)	Cr 8
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Stla Phiri
Contact Address	4724 Everlasting Daveyton
Telephone (H)	N/A
Telephone (W)	N/A

Facsimile	N/A
Cellular	062 560 7912
E-mail	Sslaphiri12@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

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LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	

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Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1st Reassessment	Assessors summary report / instruments	Learner file Date:	
2nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place	Eastc Electrical
Language	English
Resources	
Barriers	N/A

People to be involved with assessment

Learner: Stla Phiri	Manager:
Trainer: Miss R Nyakala	Mentor / Coach:
Assessor: Miss R Nyakala	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Stla Phiri

Learner's signature: _____ Date: _____

Assessors Name: Miss Ruth Nyaga

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
   - If the laid down assessment procedures were not followed during assessments
   - If not all evidence available was taken into account during the assessment
   - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
   - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
   - Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.

3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal "form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.

4. Should the internal moderator re-affirm the assessor' decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

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Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

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Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate **and put them at ease**	✓
2	**Explain the purpose of the meeting** (why you are there and how long the meeting will take)	✓
3	**Explain the** ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	**Explain** ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	**Explain the assessment process?** ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	**Give Learner copies of the following documentation and explain each document** ▪ The Assessment Guide which includes o The relevant unit standard (s) o Assessment contract o Assessment plan o Observation checklist o Knowledge checklist	✓
7	**Discuss the assessment plan** (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	**Tell the candidate his/her rights and responsibilities, the assessment procedures and policies** ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	**Ensure the assessment environment is appropriate** or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

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Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Stella Phiri	Signature S. Phiri
Assessors Name: Ruth Nyoka	Signature:

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INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

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LEARNING UNIT 1:

KM-02-KT01: Hand tools and power tools (50%)

Learning Outcome 1:	KT0101 Types, uses and care of hand tools
Learning Outcome 2:	KT0102 Types, uses and care of portable power tools
Learning Outcome 3:	KT0103 Types, uses and care of fixed power tools

Task	Questions Description	Marks
1	Identify hand tools and describe their uses	10

* Side cutters (Diagonal cutters) Designed specifically for cutting copper and aluminum conductors cleanly
* long-nose pliers - for reaching into tight spaces, bending wires loops for terminal connections and holding small components.
* Wire Strippers: Precision tools used to remove the insulation from conductors
* Flat-head and Phillips - Used to secure terminal screws in distribution boards, light switches and socket outlets
* Spirit level: Ensures that distribution boards switches and conduits ar installed perfectly plumb vertical or level (horizontal)
* Steel rule: Used to precise measurement when marking out holes for mounting component in cabinets
* Hacksaw: Used for cutting metal conduit, trunking and cable tray to the required lengths
* Pliers, gripping, bending, cutting wires
* Tape measure - is used to measure length, width and distance accurately during installation work
* Utility knife - used to cut insulation, cable, plastic or other materials

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LEARNING UNIT 2:

KM-02-KT02: Measuring and testing instruments (50%)

Learning Outcome 1:	KT0201 Portable electrical measuring instruments
Learning Outcome 2:	KT0202 Types of electrical measuring and testing instruments
Learning Outcome 3:	KT0203 Safe use of measuring and testing instruments
Learning Outcome 4:	KT0204 Construction and operating principles of measuring and testing instruments
Learning Outcome 5:	KT0205 Methods of connecting measuring and testing instruments in circuits
Learning Outcome 6:	KT0206 Applications and methods of using electrical measuring and testing instruments
Learning Outcome 7:	KT0207 Care and maintenance of single and three phase measuring and testing instruments
Learning Outcome 8:	KT0208 Instrument transformers
Learning Outcome 9:	KT0209 Types and functions of panel mounted electrical measuring and testing instruments
Learning Outcome 10:	KT0210 Safe use of panel mounted measuring and test instruments
Learning Outcome 11:	KT0211 Protection of measuring instruments when connected in a circuit.

Task	Questions Description	Marks
2	a) Identify the basic components of various measuring and test instruments and describe the basic principle of operation. b) Describe the applications and methods of using electrical measuring and testing instruments in direct and alternating current circuits.	10

Instrument	Measures	Basic principle
Ammeter	Current (A)	Magnetic effect of current
Voltmeter ~~Voltage~~	Voltage (V)	Electromagnetic deflection
Ohmmeter	Resistance (Ω)	Ohm's law
Multimeter	V, A, Ω	Electronic measurement circuit
Clamp meter	Current	Electromagnetic induction
Megger	insulation resistance	High voltage resistance test

2 b.

instrument	Application	Connection method
Ammeter	measure current	Connected in series
~~Voltage~~ Voltmeter	measure voltage	Connected in parallel
Ohmmeter	measure resistance	Circuit must be off
multimeter	Measure V, A, Ω	Series or parallel depending on measurement
Clamp meter	Measure current in live wires	Clamp around one conductor
Megger	Test insulation resistance	Connected between conductor and Earth

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Task	Questions Description	Marks
3	Describe safety and functionality checks to be performed on measuring and testing instruments before use	5

*Before using any electrical measuring or testing instrument you must perform specific safety and functionality checks to ensure the accuracy of your reading and most importantly your personal safety

* Visual Safety inspection
- Check the instrument Body
- Check inspect test leads and Probes
- Terminal integrity
* Battery and power status
- Battery Check
- Power on self-test (POST)
* Functionality and accuracy verification
- the proof of function
- test on known load
- Calibration verification
* Setting Selection
- Correct Range
- Terminal Selation

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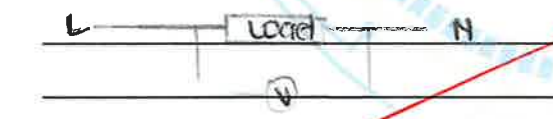
Task	Questions Description	Marks
4	a) Describe correct methods of handling and storing measuring and testing instruments b) Describe, with the aid of drawings, the connection of measuring instruments to single phase and three phase circuits	10

Step	Action ~~Why~~	Why
Pre - Check	inspect leads / body for damage	insures personal safety
Function	verify on a known live source	Ensure accurate readings
Usage	use proper ports and ranges	Prevents equipment damage
Storage	store in a dry, dedicated case	Preserves calibration and lifespan

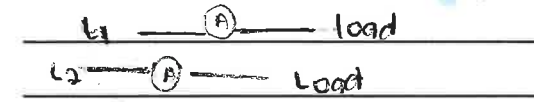
b. Ammeter in a Single - phase Circuit



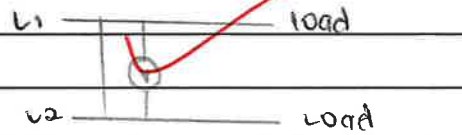
voltmeter in a Single - phase



3. Ammeter in three - phase

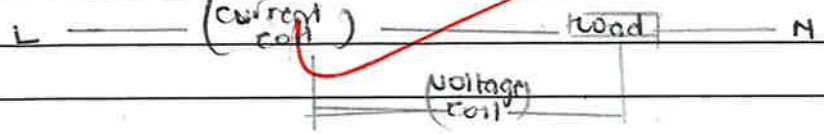


voltmeter in 3 phase

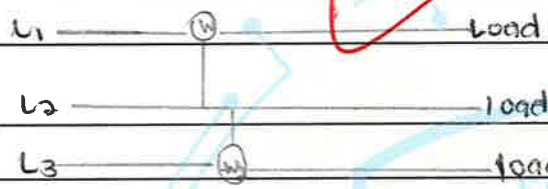


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Wattmeter in a Single phase



wattmeter in the three phase



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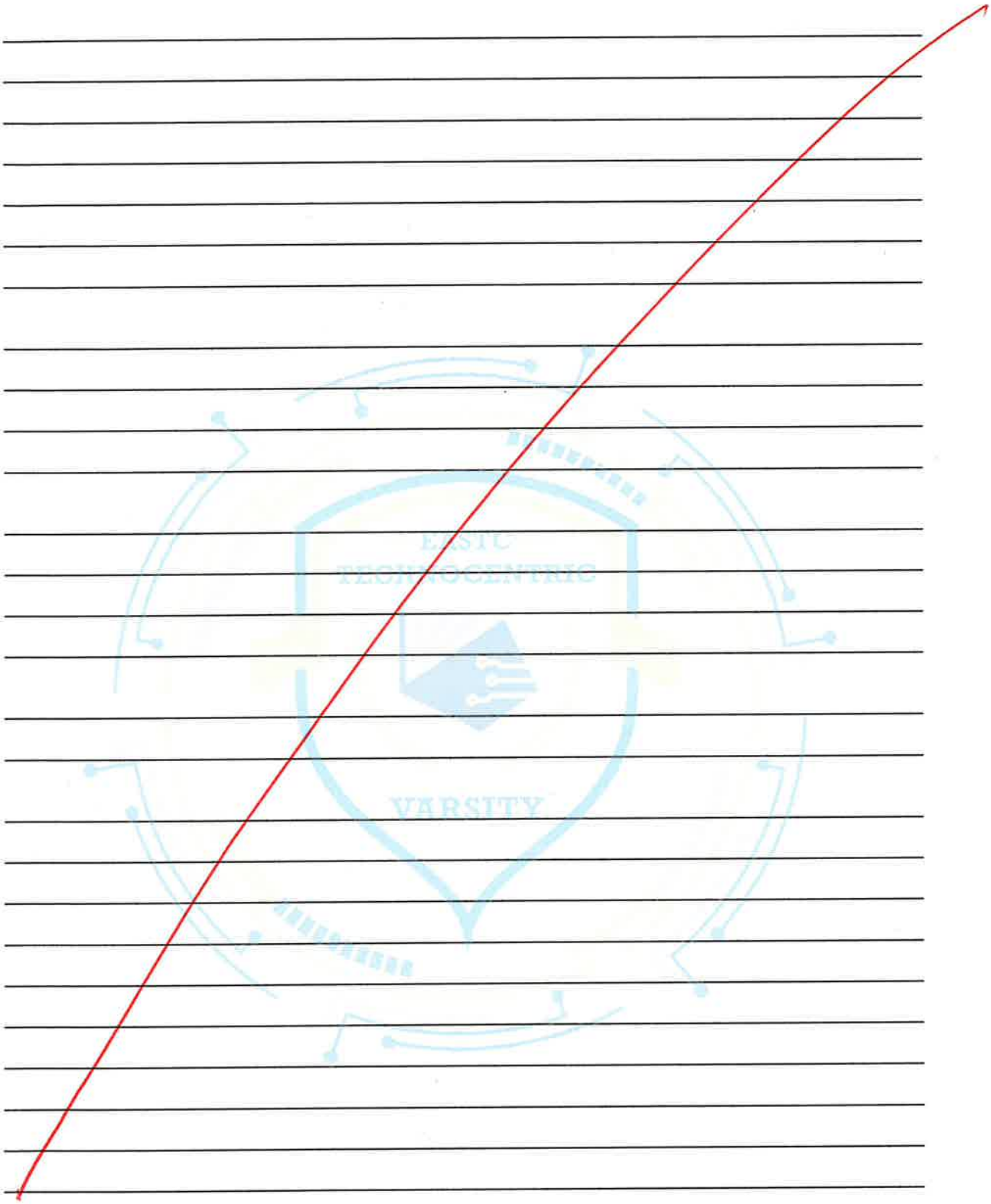
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Task	Questions Description	Marks
5	Describe the various methods of protection when connecting measuring instruments in circuits	10

- A fuse is installed inside many measuring instrument to protect them from excess current
- If too much current flowe through the instrument the fuse melts and breaks the circuit
- Shunt Resistor Protection is connected in parallel wit the ammeter
- it protect it allows most of the current to bypass the meter movement
- Series resistor - limits current in voltmeter
- Correct range selection - Prevents instrument overload
- Proper connection - Avoids Shorts circuits
- insulation and covers - Protects user from electrica Shock
- Clamp meter use - Allows safe current measuremend
- Isolation Transformer - isolation transformer Separates the measuring instrument from the main power supply
- Proper range selection - protects the user from electric Shock
- Prevent short circuits in the circuit

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Assessor's Report

Candidate's Name	Sela Phiri	ID No.	940611175089
Assessor's Name	RUTH NYAKALA	Reg. No.	
Title	Tools, equipment and materials		

ASSESSMENT DECISION

Outcome	C	NYC	Comments

OVERALL COMPETENCE JUDGEMENT:

Comments

Date	
Signature of Assessor	Signature of Candidate S. Phiri

Candidate Feedback Report

Candidate's Name	Sela Phiri	ID No.	940611175089
Assessor's Name	Ruth Nyakala	Reg. No.	
Title	Tools, equipment and materials		

ASSESSMENT DECISION

Source of Evidence	C	NYC	Comment
Knowledge questions			
Product /Work sample			
Project			
Indirect Evidence			
Overall Assessment Decision			
Additional Notes			
Date			
Signature of Assessor		Signature of Candidate	S.

S

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

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Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Sela Phin	ID No.	940611175089
Title	Tools, equipment and Materials		

MODERATION DECISION

Outcome	C	NYC	Comments
Overall Moderation Decision			

Feedback to Assessor	
Action Required	
Date of Moderation	
Signature of Moderator	
Signature of Assessor	
Signature of Candidate	S. Phin

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ASSESSMENT REVIEW

NAME of LEARNER Seta Phiri	NAME of ASSESSOR
VENUE Eastc	DATE of REVIEW

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment related to the registered unit standard?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment was practical?	☐Agree ☐Disagree	☐Agree ☐Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment instruments were fair, clear and understandable	☐Agree ☐Disagree	☐Agree ☐Disagree	
The assessment judgements was made against set requirements	☐Agree ☐Disagree	☐Agree ☐Disagree	
The venue and equipment was functional?	☐Agree ☐Disagree	☐Agree ☐Disagree	
Special needs were identified and the assessment plan was adjusted	☐Agree ☐Disagree	☐Agree ☐Disagree	
Feedback was constructive against the evidence required	☐Agree ☐Disagree	☐Agree ☐Disagree	
An opportunity to appeal was given	☐Agree ☐Disagree	☐Agree ☐Disagree	
The evidence was recorded	☐Agree ☐Disagree	☐Agree ☐Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING

I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid

Learner Date	Assessor Date	Moderator Date
S.		

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